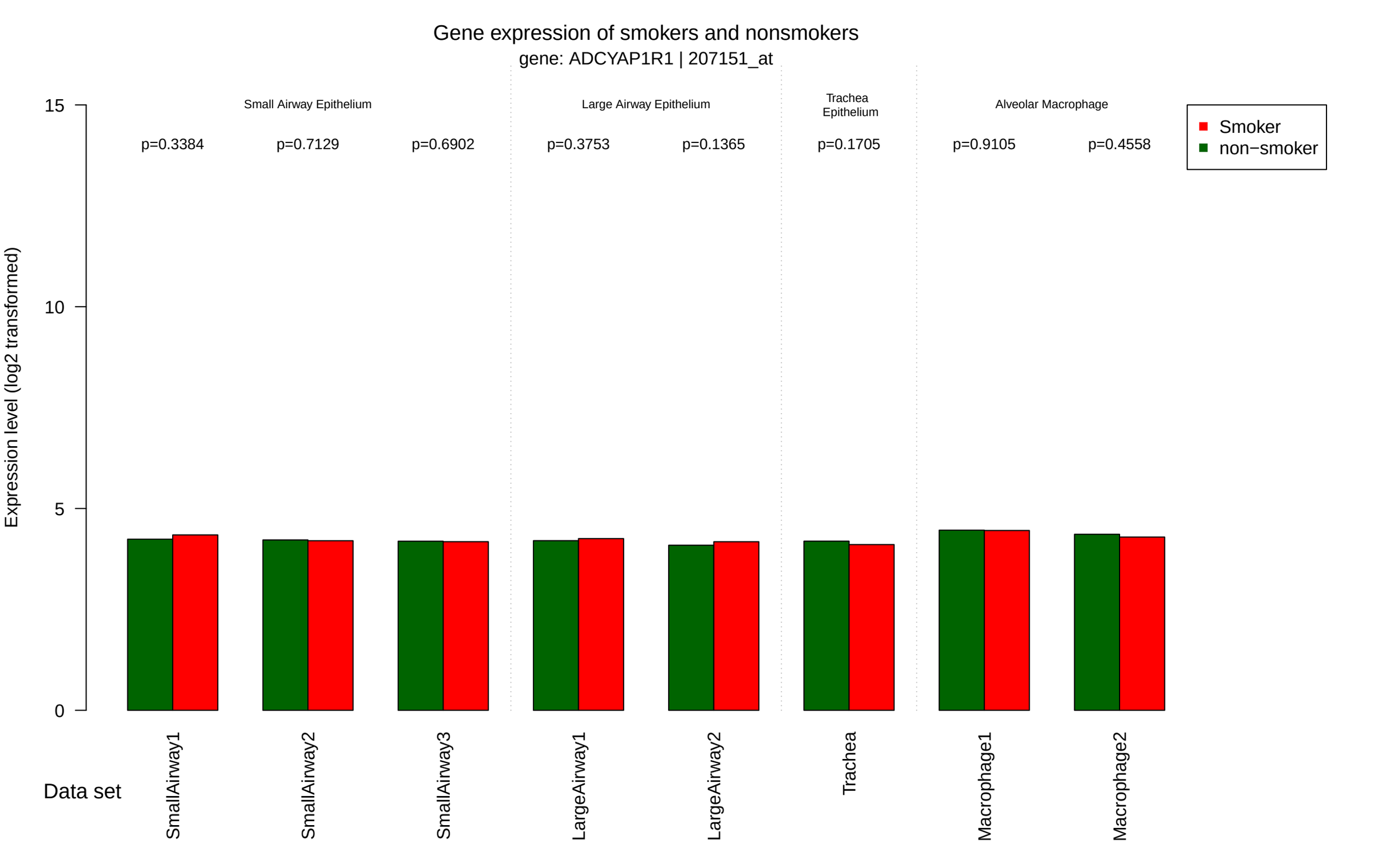
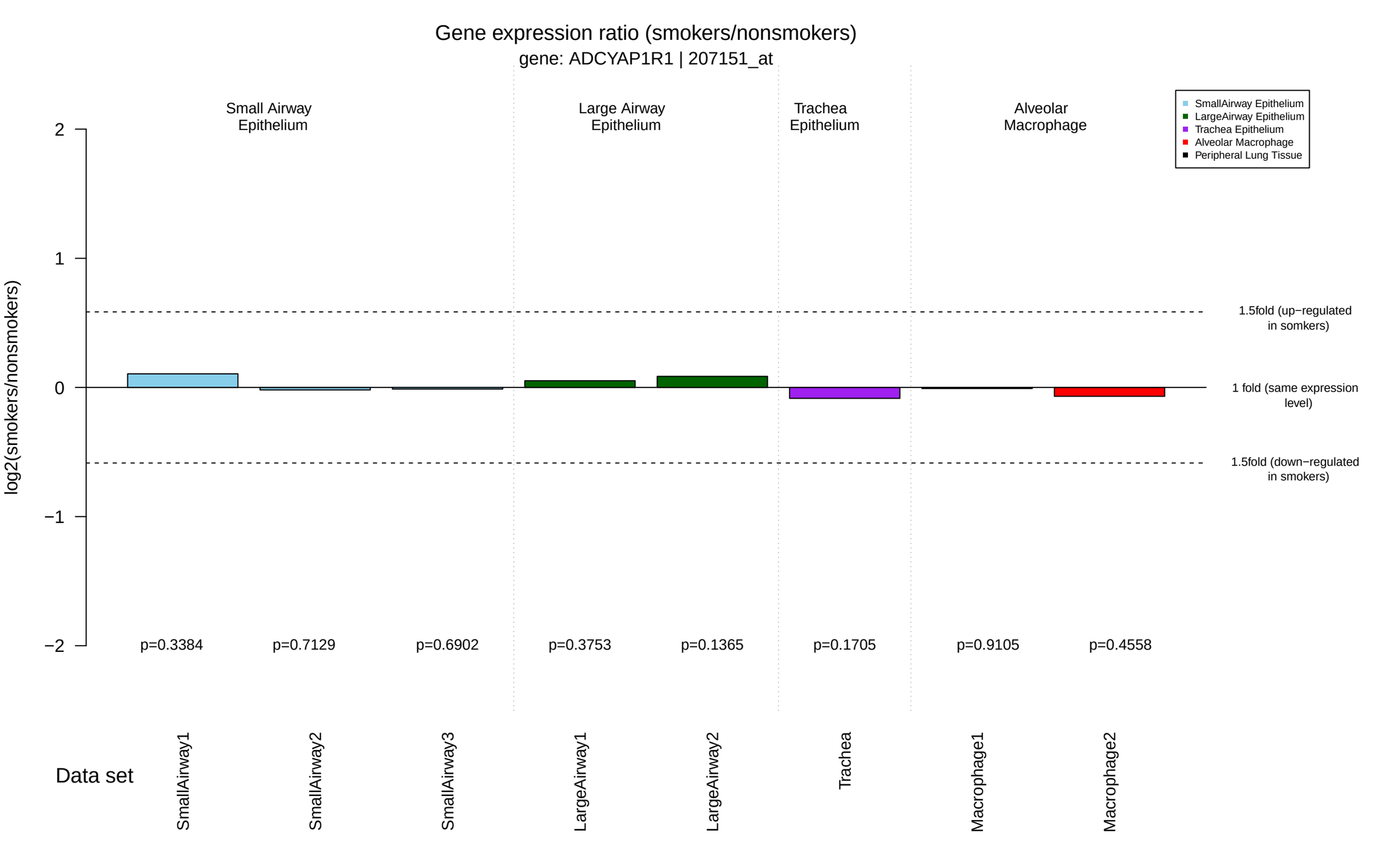
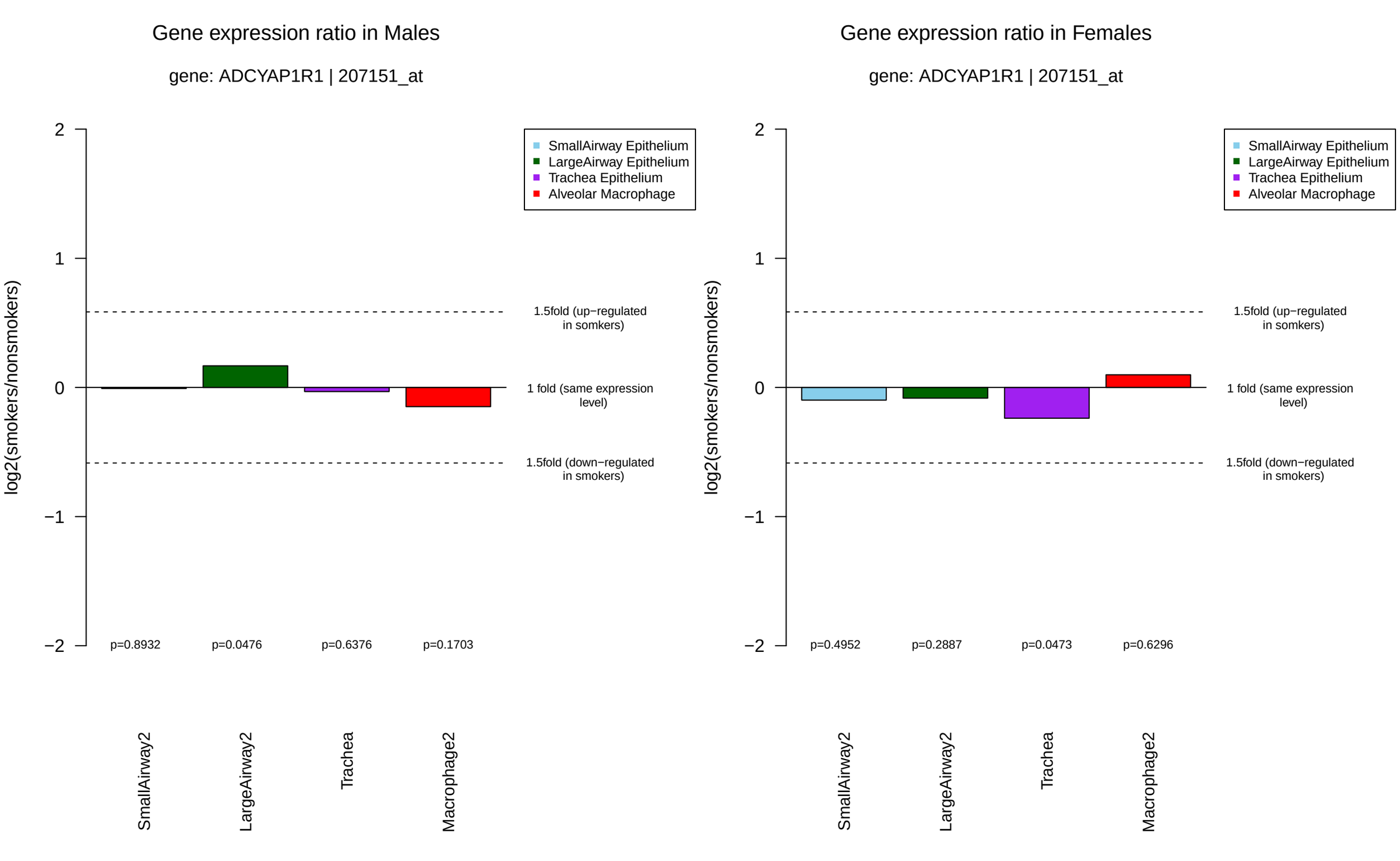
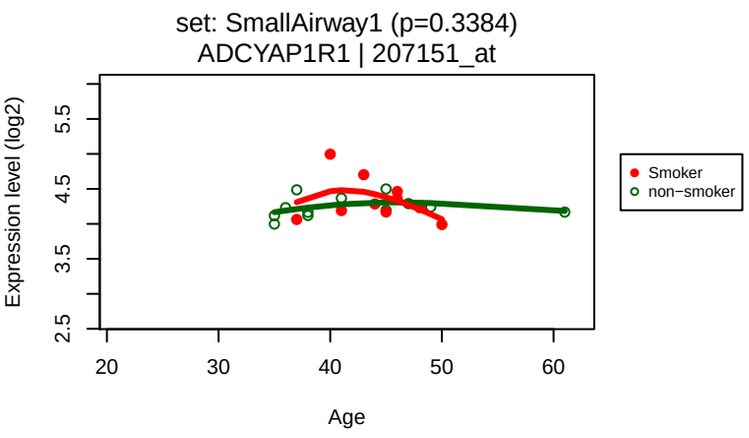
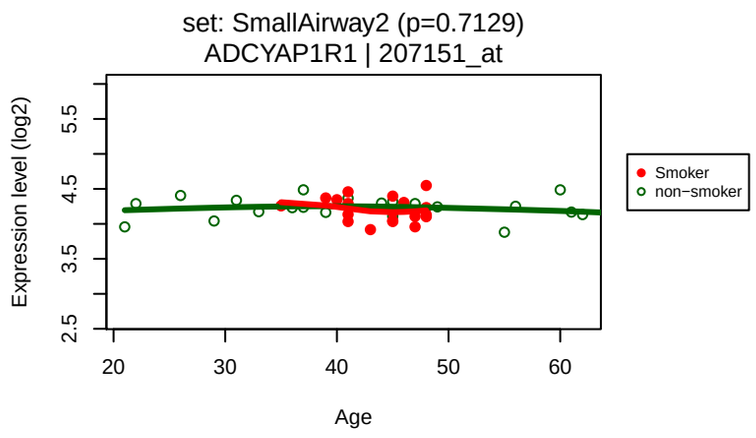
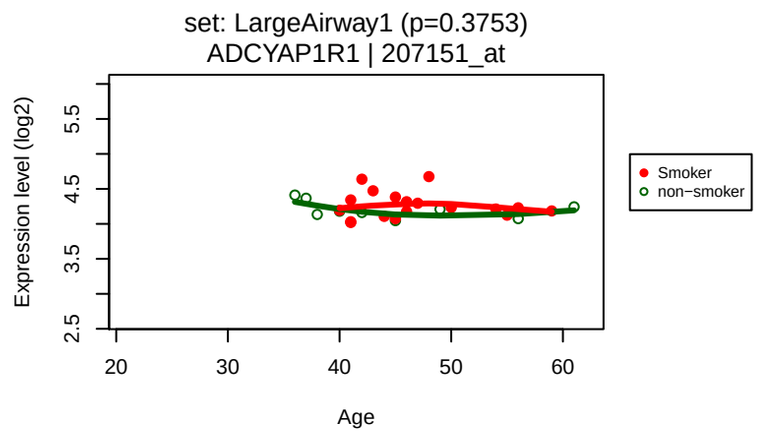

Gene expression of smokers and nonsmokers
gene: ADCYAP1R1 | 207151_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3384
p=0.7129
p=0.6902
p=0.3753
p=0.1365
p=0.1705
p=0.9105
p=0.4558
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

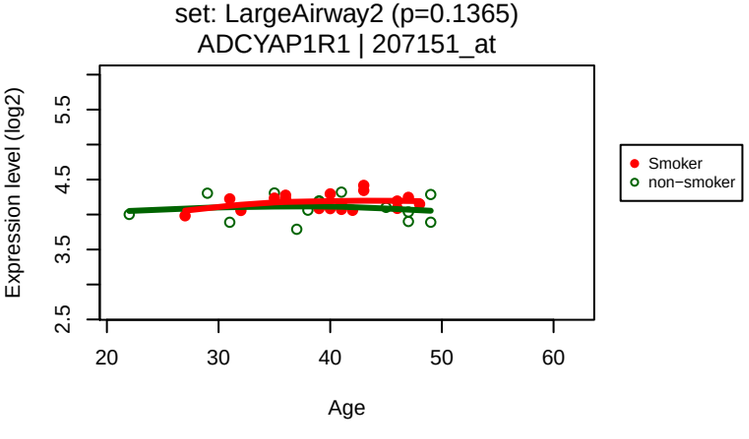

Gene expression ratio (smokers/nonsmokers)
gene: ADCYAP1R1 | 207151_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3384
p=0.7129
p=0.6902
p=0.3753
p=0.1365
p=0.1705
p=0.9105
p=0.4558
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

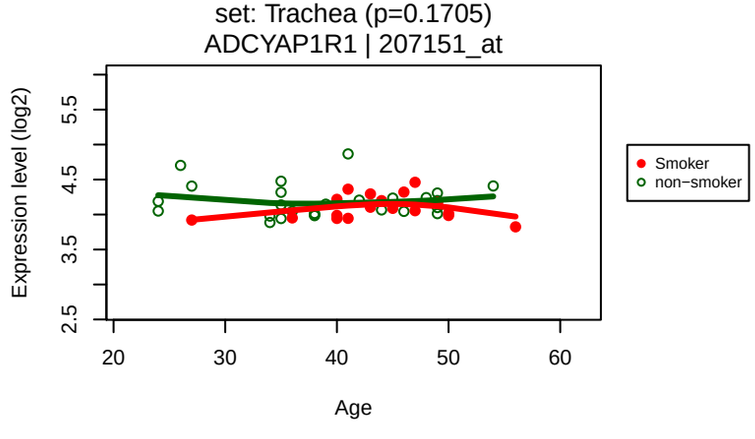

Gene expression ratio in Males
gene: ADCYAP1R1 | 207151_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8932
p=0.0476
p=0.6376
p=0.1703
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCYAP1R1 | 207151_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4952
p=0.2887
p=0.0473
p=0.6296
SmallAirway2
LargeAirway2
Trachea
Macrophage2

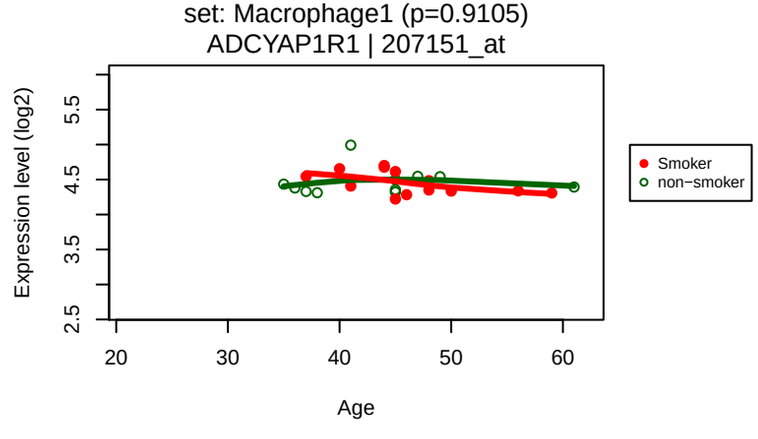

set: SmallAirway1 (p=0.3384)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

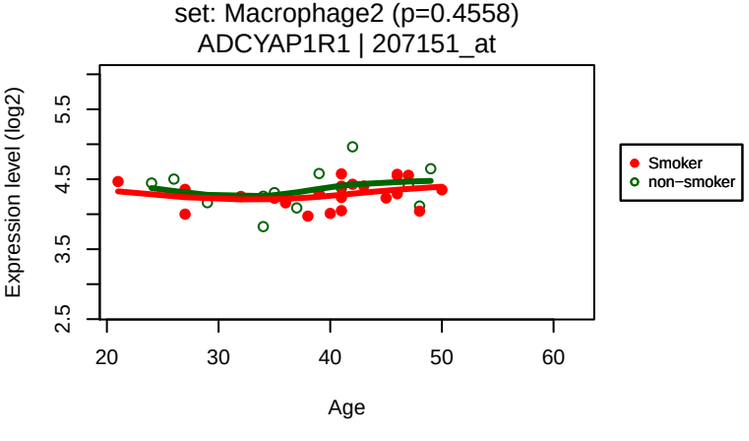

set: SmallAirway2 (p=0.7129)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

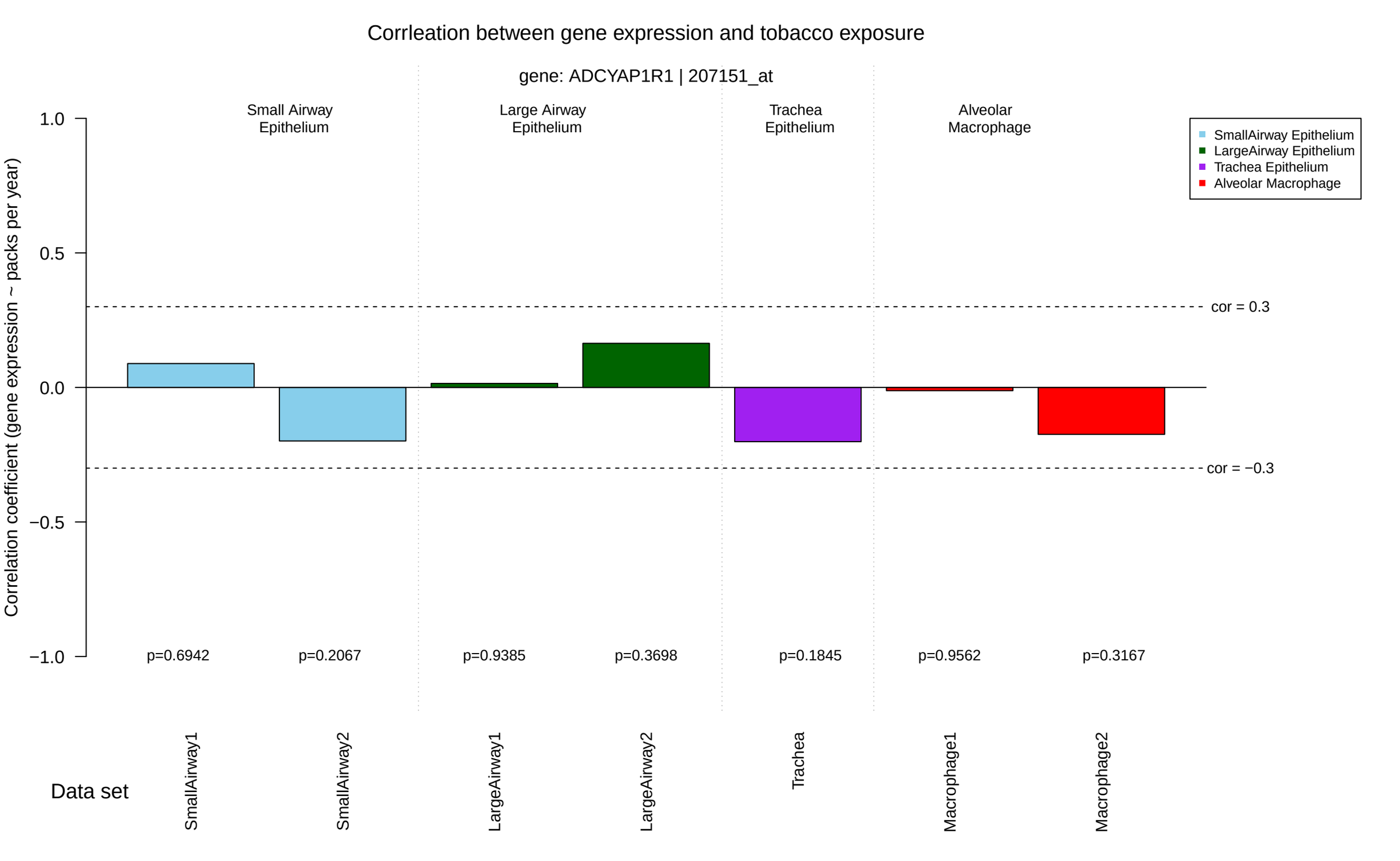

set: LargeAirway1 (p=0.3753)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1365)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1705)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9105)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4558)
ADCYAP1R1 | 207151_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCYAP1R1 | 207151_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6942
p=0.2067
p=0.9385
p=0.3698
p=0.1845
p=0.9562
p=0.3167
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2